



## Data Science and AI



### Differentiation

Data driven AI systems that automatically prepare or make operational decisions.



### Situation

Predictive AI systems fully automate operational decisions such as staffing, pricing, or resource allocation.

### Leadership Influence

Leadership bears responsibility for decisions that follow optimization logics rather than individual judgment.



## Immersive Technologies



### Differentiation

Digital technologies that permanently shift work and interaction spaces into virtual or mixed reality environments.



### Situation

Teams work, communicate, and make decisions primarily through avatars. Some employees feel alienated, while others thrive.

### Leadership Influence

Leadership must establish cohesion, trust, and orientation without physical presence and actively shape how permanent immersion is handled.

## Digital Twins of Humans



### Differentiation

AI representations of real individuals that are based on their decision making and behavioral patterns and act on their behalf.



### Situation

Leaders have digital twins that conduct meetings and make decisions. It becomes unclear whether the human or their digital twin is acting.

### Leadership Influence

Leadership must define where delegation ends and personal responsibility begins, and how authenticity and trust are to be preserved.

## Sensor Technologies



### Differentiation

Ubiquitous sensors that continuously collect behavioral, performance, and health data from people and their environments.



### Situation

Sensors in clothing, buildings, and bodies provide real time data on productivity and health. Many employees perceive this as constant surveillance.

### Leadership Influence

Leadership must find a balance between data driven control, risk prevention, and the protection of privacy and trust.



## Generative AI 2.0

### Differentiation

AI systems that autonomously generate creative, strategic, or communicative content.



### Situation

Creative processes are dominated by multimodal generative AI. This raises the question of whether human creativity is still needed.

### Leadership Influence

Leadership must decide where human creativity is complemented, replaced, or redefined.



## Autonomous Robots

### Differentiation

Physical, autonomous systems that independently perform tasks and make context dependent decisions.



### Situation

Autonomous robots take over complex tasks and work side by side with humans. They make local decisions without direct human control.

### Leadership Influence

Leadership must develop new role models for human machine collaboration and actively address fears of job loss.

## Brain-Computer Interfaces (BCI)

### Differentiation

Neurotechnological interfaces that enable direct interaction between the brain and digital systems.



### Situation

BCIs are deployed organization wide in safety critical and time critical contexts. Decision making and communication are accelerated through neural signals.

### Leadership Influence

Leadership must clarify where cognitive support ends and responsibility begins, and who decides on the use and limits of this technology.

## Human Enhancement

### Differentiation

Biological and neurotechnological interventions aimed at the permanent enhancement of human performance.



### Situation

Genetic modification and neurotechnological implants for performance enhancement are widespread. This can lead to tensions between employees with and without performance enhancements.

### Leadership Influence

Leadership must decide which performance standards apply and where the boundaries lie between voluntariness, expectation pressure, and fairness.